Haruna Tomono

linkedin.com/in/harunatomono | github.com/harunatomono

EDUCATION

University of California, San Francisco 2026 - Present
Ph.D. in Biological and Medical Informatics *San Francsico, CA*

California Institute of Technology (Caltech) 2020 - 2024
B.S. in Bioengineering, Minor in Neurobiology *Pasadena, CA*

RESEARCH EXPERIENCE

Broad Institute of MIT and Harvard / Dana-Farber Cancer Institute 2024 - 2026
Principal Investigators: Dr. Rameen Beroukhim & Dr. Matthew Meyerson *Cambridge, MA*

- Analyzed and visualized copy number alterations in TCGA Pan-Cancer Atlas; established quality metrics and consensus on copy number and purity-ploidy modeling with external collaborators
- Integrated RNA-seq, deep mutational scanning, multi-omic, and WGS data to identify mechanisms of sensitivity and resistance to sevabertinib; collaborated with experimental colleagues and industry team (Siegel et al., 2025)
- Re-engineered copy-number calling pipelines for expanded data compatibility for hybrid-capture targeted sequencing to analyze 10,000+ understudied samples

Caltech, Chou Lab 2021 - 2024
Principal Investigator: Dr. Tsui-Fen Chou *Pasadena, CA*

- Designed and validated a dual-tag immunoprecipitation mass spectrometry protocol to track protein trafficking across organelles
- Characterized mito-nuclear protein trafficking using a novel dual-tag MS method to elucidate Alzheimer's disease-related pathways
- Explored proteogenomic characterization of rainbow mantis shrimp ocular proteins

Calico Life Sciences, Proteomics 2023
Principal Investigator: Dr. Fiona McAllister *South San Francisco, CA*

- Optimized a novel, multiplexed sample preparation method to increase throughput/reproducibility for native protein quantification
- Profiled immune pathways in aging mouse kidney subpopulations with laser-capture microdissection LC-MS, collaborating with the Microscopy, Genomics, and Metabolomics groups (Chan et al., 2024)
- Standardized immunofluorescence imaging-coupled MS workflows for downstream analysis

Regeneron Pharmaceuticals, Analytical Chemistry 2022
Supervisor: Dr. Bo Zhao *Tarrytown, NY*

- Co-invented a novel, streamlined antibody-depletion protocol for host cell protein detection, resulting in a first-author publication and patent application (Tomono et al., 2025; patent pending)
- Benchmarked protein identification across methods, providing actionable recommendations for drug production improvement

PATENTS AND PUBLICATIONS

Siegel, F., Siegel, S., Kotýnková, K., Uzunba, G. K., Korr, D., **Tomono, H.**, … & Greulich, H. (2025). Sevabertinib, a Reversible HER2 Inhibitor with Activity in Lung Cancer. *Cancer Discovery*. 10.1158/2159-8290.CD-25-0605

Tomono, H., Su, Y., Adaniya, S., Zhao, B., Xiao, H., & Li, N. (2025). Host cell protein detection via surfactant-assisted acid precipitation. *Journal of Pharmaceutical Sciences*, 114(11), 104008. 10.1016/j.xphs.2025.104008

Chan, L. J., Olsson, N., Preciado López, M., Hake, K., **Tomono, H.**, Veras, M. A., & McAllister, F. E. (2024). Plasma and kidney proteome profiling combined with laser capture microdissection reveal large increases in immunoglobulins with age. *Proteomes*, 12(2), 16. 10.3390/proteomes12020016

Xiao, H., Li, N., Zhao, B., **Tomono, H.**, Hu, M., Molden, R., Hu, Y., Huang, Y., & Qiu, H. (patent pending). Size exclusion chromatography for characterizing host cell proteins. *Regeneron Pharmaceuticals, Inc.* U.S. Patent Application 20230279046.

PRESENTATIONS

Tomono, H., ... & Beroukhim, R. (2026, April 17-22). *Copy number analysis of regulatory regions reveal recurrent promoter/enhancer somatic copy number alterations across >8,000 TCGA samples* [Poster presentation]. AACR Annual Meeting 2026, San Diego, CA.

Siegel, F., Siegel, S., Kotýnková, K., Uzunbas, G. K., Korr, D., **Tomono, H.**, ... & Greulich, H. (2025, October 22-26). *Sevabertinib, a Reversible HER2 Inhibitor with Activity in Lung Cancer* [Poster presentation]. Presented by Greulich, H., AACR-NCI-EORTC International Conference on Molecular Targets and Cancer Therapeutics, Boston, MA.

Tomono, H., ... & Greulich, H. (2025, October 22-26). *Identifying mechanisms of sensitivity and resistance to sevabertinib* [Oral presentation]. Dana-Farber Cancer Institute Center for Cancer Genomics Meeting, Boston, MA.

Siegel, F., Kotýnková, K., Uzunbas, G. K., Jourdan, T., Berthold, M., Siegel, S., Korr, D., Berger, M., **Tomono, H.**, ... & Greulich, H. (2025, April 25-30). *BAY 2927088: a novel treatment option for patients with NSCLC with HER2 mutations* [Poster presentation]. Presented by Siegel, F., American Association for Cancer Research Annual Meeting, Chicago, IL.

Uzunbas, G.K., Chou, A., Gupta, A., **Tomono, H.**, ... & Greulich, H. (2024 December 16-17). *Deep mutational scanning identifies resistance mutations for EGFR/ERBB2 exon 20 insertion inhibitor, BAY 2927088* [Poster presentation]. Presented with Chou, A., Broad Institute Annual Scientific Retreat, Boston, MA.

Tomono, H., ... & McAllister, F. (2023, August 17-18). *Laser-capture microdissection coupled with mass spectrometry for high-throughput protein and protein complex analysis* [Poster presentation]. Calico Life Sciences Research Symposium, South San Francisco, CA.

Tomono, H., ... & Li, N. (2022, August 24). *Method development for host cell protein identification via surfactant-aided acid precipitation* [Oral presentation]. Regeneron Pharmaceuticals Analytical Chemistry Meeting, Tarrytown, NY.

Tomono, H., ... & Chou, T.F. (2021, August 19). *Developing a dual-tag mass spectrometry technique to examine directional protein trafficking between the mitochondria and nucleus* [Oral virtual presentation]. Caltech Summer Undergraduate Research Seminar Day, Pasadena, CA.

ACTIVITIES AND LEADERSHIP

Caltech Student Wellness Services 2022 - 2024
Senior Health Advocate (*American Red Cross Emergency Medical Responder*)

- Stabilized patients, coordinated with EMS/ER staff, and filed follow-up reports
- Mentored new health advocates and conducted peer training sessions for incoming cohorts

Caltech Office of Undergraduate Admissions 2022 - 2024
Admissions Ambassador

- Presented at student panels on undergraduate research experience
- Led ∼100 campus tours, highlighting academic programs, research opportunities, and student culture

Caltech Undergraduate Student Leadership 2023 - 2024
Avery House Senior Representative

- Negotiated building access and safety compliance between student leaders and university administration; resolved student conflicts as a liaison between peers and administration
- Coordinated campus-wide logistics and safety planning for Caltech's largest undergraduate tradition, Ditch Day

Caltech Bioengineering Student Faculty Committee 2023
Undergraduate Representative

- Synthesized student feedback on major curriculum into actionable proposals with recommendations to faculty, leading to 4 additional elective options and resolving 3+ common course conflicts
- Presented alongside professors and major representative to evaluate and improve requirements, courses, and departmental resources

TECHNICAL SKILLS

Programming: Python, R, bash scripting, MATLAB, Mathematica, Git/GitHub

Bioinformatics & Data Analysis: RNA-seq & WGS pipelines, differential expression, copy-number calling

Experimental methods: Mass spectrometry (label-free, isobaric, co-fractionation), Western blotting, chromatography (size exclusion, liquid), mammalian cell culture, immunoprecipitation